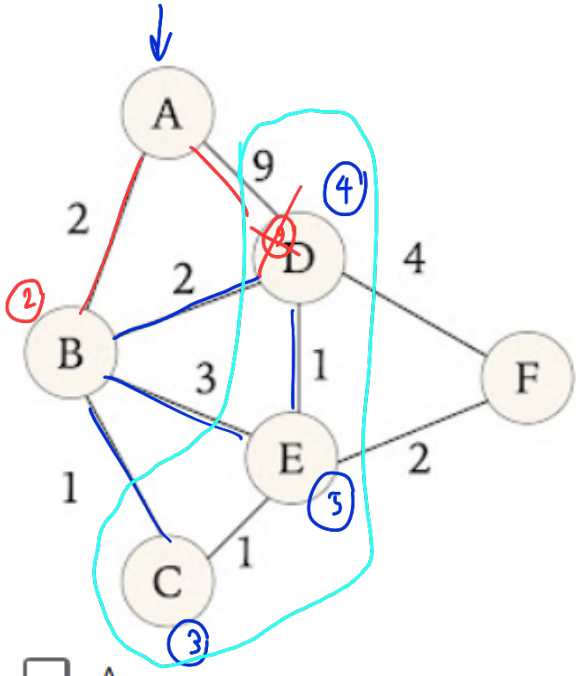
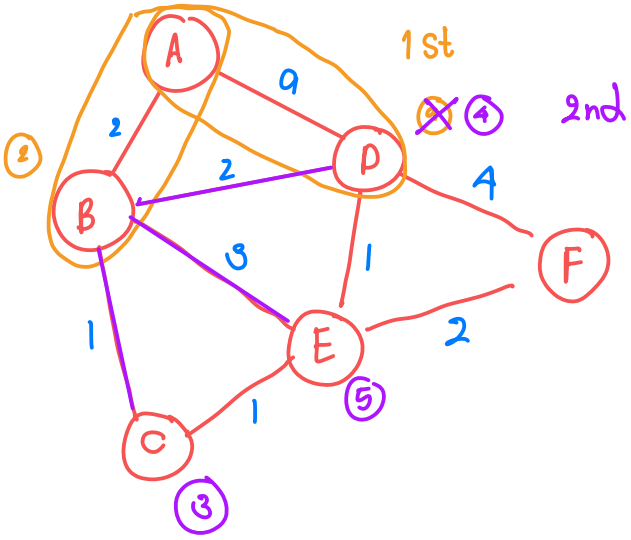
ข้อสอบปีที่แล้ว

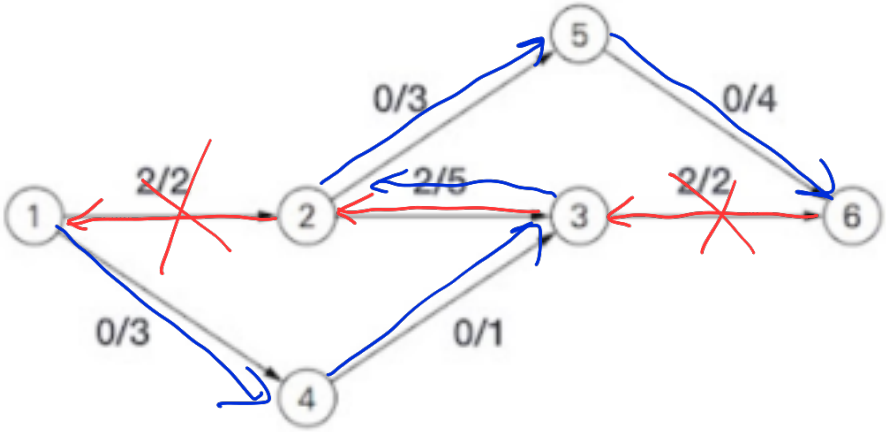
From the graph below, if we apply Dijkstra's algorithm with node A as the source node, the labels of which nodes will change in the 2nd iteration when the edge relaxation is performed ?



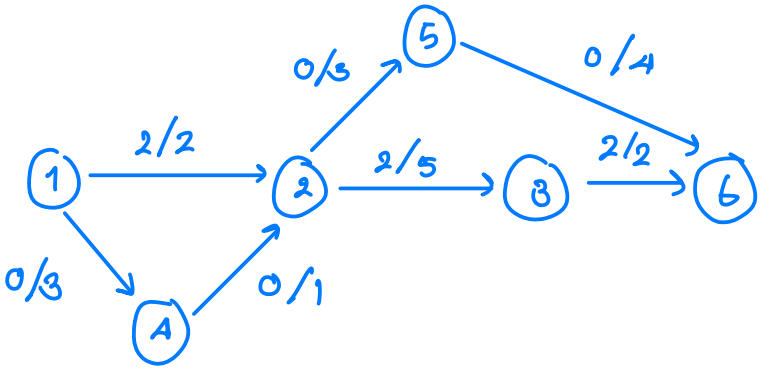


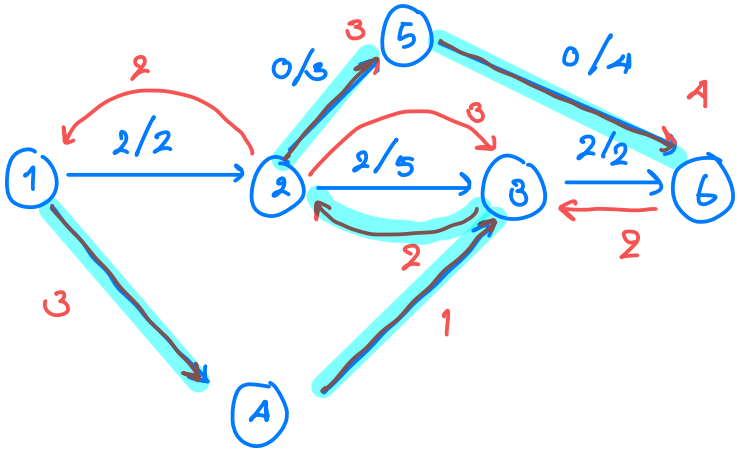
- [ ] A
- [ ] B
- [ ] C
- [ ] D
- [ ] E
- [ ] F

For the following flow network with source node 1 and destination node 6, which one is a valid augmenting path in the residual graph ?



- 1 -> 2 -> 3 -> 4
- 1 -> 4 -> 3 -> 2 -> 5 -> 6
- 1-> 2 -> 5 -> 6
- 1 -> 4 -> 3 -> 6

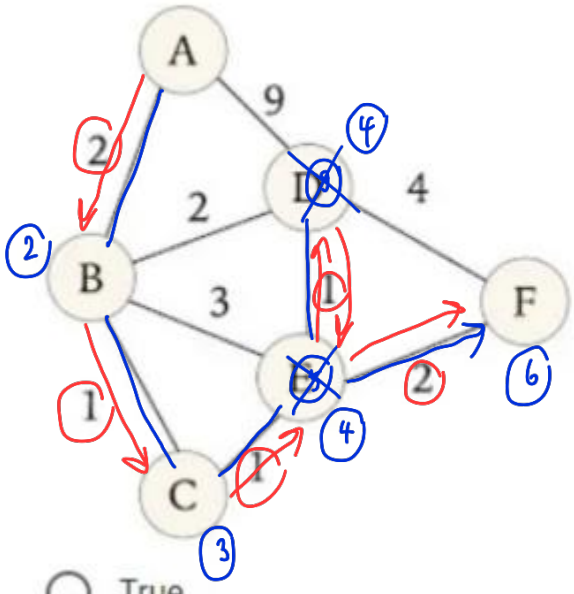




Prims, kruskal

From the graph below, the minimum spanning tree is the same as the shortest path tree rooted at node A.

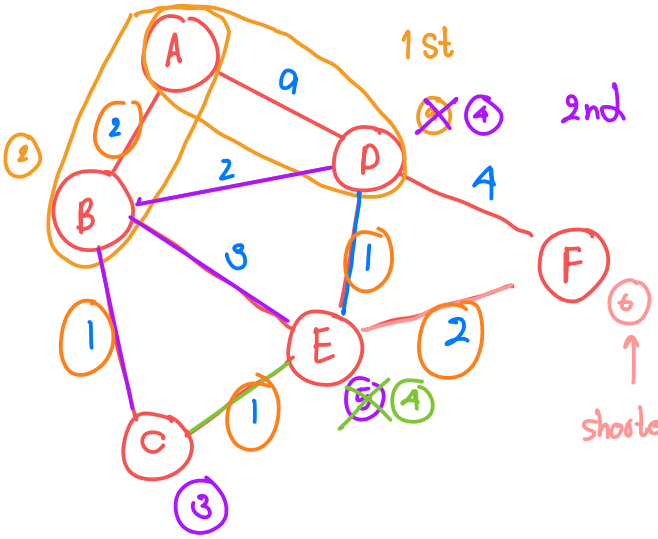
Dijkstra



path 6

Span 7

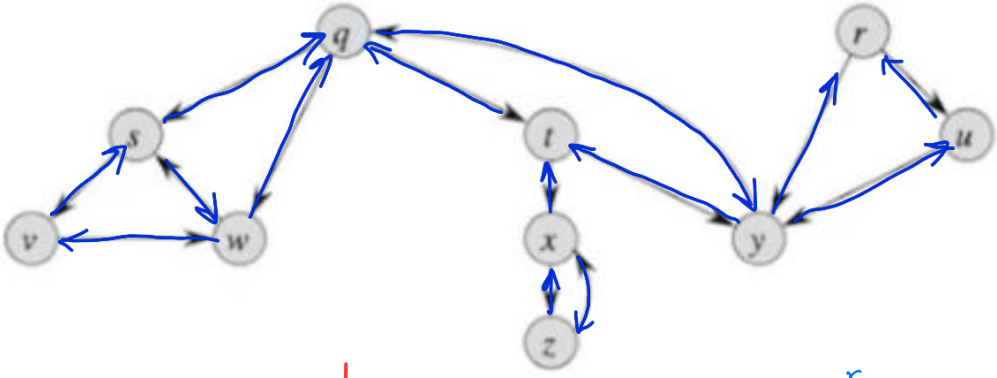
- True
- False



shortest path = 6

Spanning = 7

From the graph below, suppose we run Kosaraju's algorithm starting at node t (break ties alphabetically), node w will be in which SCC based on the order of SCCs produced by the algorithm?



- First SSC
- Second SSC
- Third SSC
- Fourth SSC
- Fifth SSC

r
u
y
q
t
s
v
w
x
z

r 1 SCC

u 2 SCC

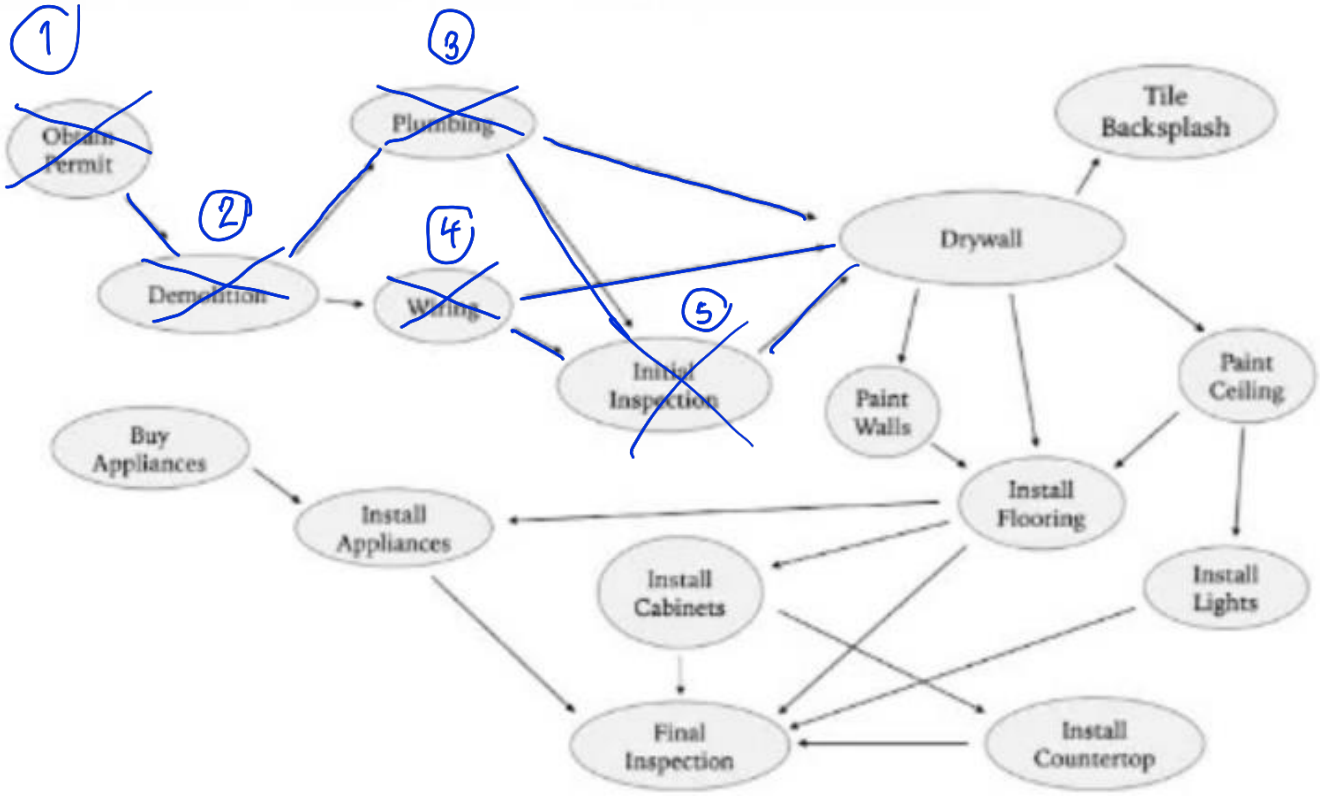
y → t → q 3 SCC

s → w → v 4 SCC

x → z 5 SCC

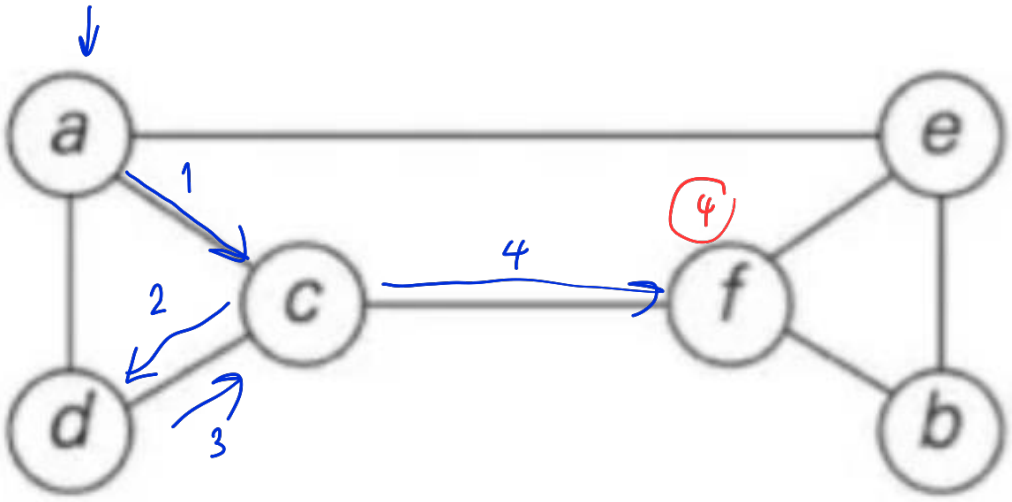
$r_{4,1}$ $y_{3,3}$ $q_{2,4}$ t_1 $u_{5,2}$ v_8 w_7 s_6

Given the task dependencies of kitchen remodeling project below, which task below cannot be done within the first five orders based on the topological order produced by the source removal algorithm? * 1 คะแนน



- ○ Plumbing
- ◉ Drywall
- ○ Buy Appliances
- ○ Obtain Permit
- ○ Wiring
- ○ Initial inspection

If we run Depth first search by starting at node A, and then the subsequent nodes are visited alphabetically, what is the order at which node F is visited?



- 2
- 3
- 4
- 5
- 6

$d_{3,1}$ $f_{4,}$
$c_{2,}$
$a_{1,}$

Which abstract data structure is used during an execution of Prim's algorithm to determine the next node to be added to the tree?

- ○ List
- ◉ Priority queue
- ○ Dictionary
- ○ Singly-linked list
- ○ Binary search tree

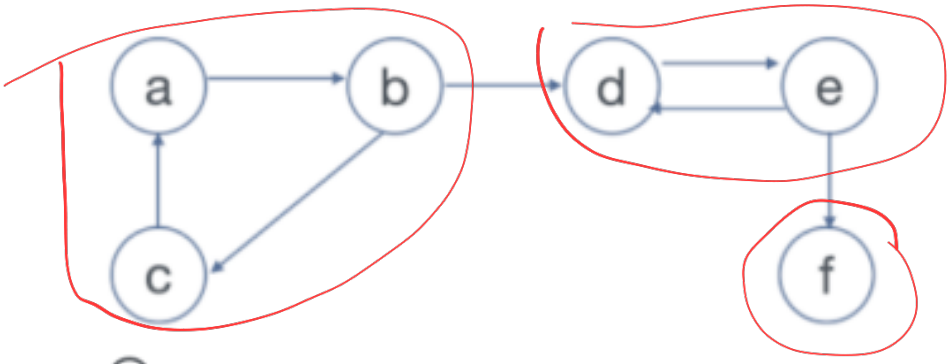
An edge in a flow network has the edge capacity of 16 units and a flow of 5 units. What are the capacities of the forward edge and the reverse edge in the residual graph ?

- ○ Forward edge capacity of 5 units and Reverse edge capacity of 5 units
- ○ Forward edge capacity of 16 units and Reverse edge capacity of 11 units
- ◉ Forward edge capacity of 11 units and Reverse edge capacity of 5 units
- ○ Forward edge capacity of 11 units and Reverse edge capacity of 16 units
- ○ Forward edge capacity of 5 units and Reverse edge capacity of 11 units

Cycle detection is needed during an execution of Kruskal's algorithm. *

- ◉ True
- ○ False

How many strongly connected components are there in the graph below? *

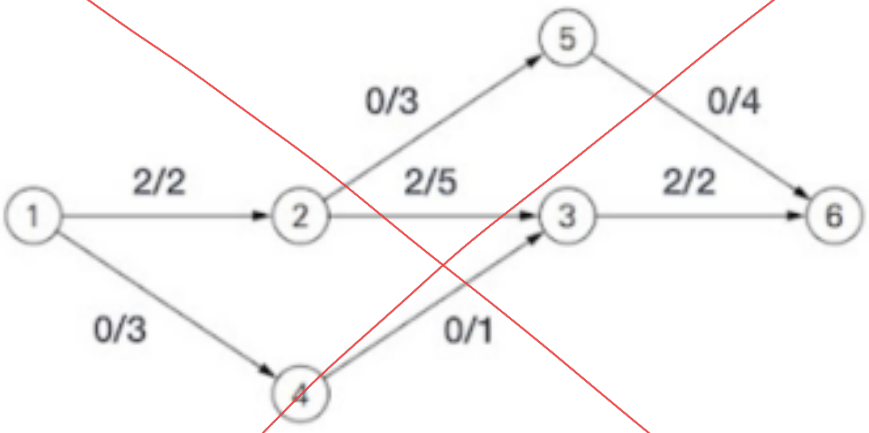


- ○ 2
- ◉ 3
- ○ 4
- ○ 5
- ○ 6

Which problem(s) below can be solved directly using a minimum spanning tree algorithm ?

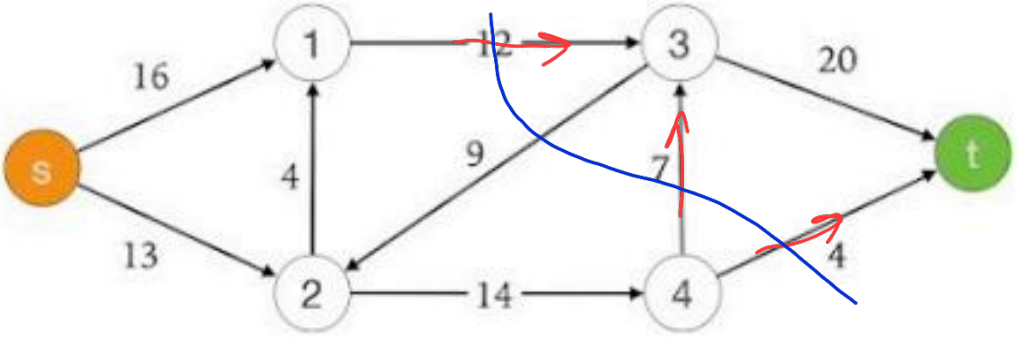
- ☑ Discovering a cluster of customers with similar characteristics.
- ☑ Finding a minimum cost path between any two nodes in a graph.
- ☐ Finding a strongly connected component in a graph.
- ☐ Designing a transmission network with smallest total link cost.
- ☐ Finding a largest cost path between any two nodes in a graph.

For the following flow network with source node 1 and destination node 6, which one is a valid augmenting path in the residual graph ?



- 1 -> 2 -> 3 -> 4
- 1 -> 4 -> 3 -> 2 -> 5 -> 6
- 1-> 2 -> 5 -> 6
- 1 -> 4 -> 3 -> 6

For the following s-t flow network, what is the minimum cut ? *



X = {s, 1} 13 + 12 = 25

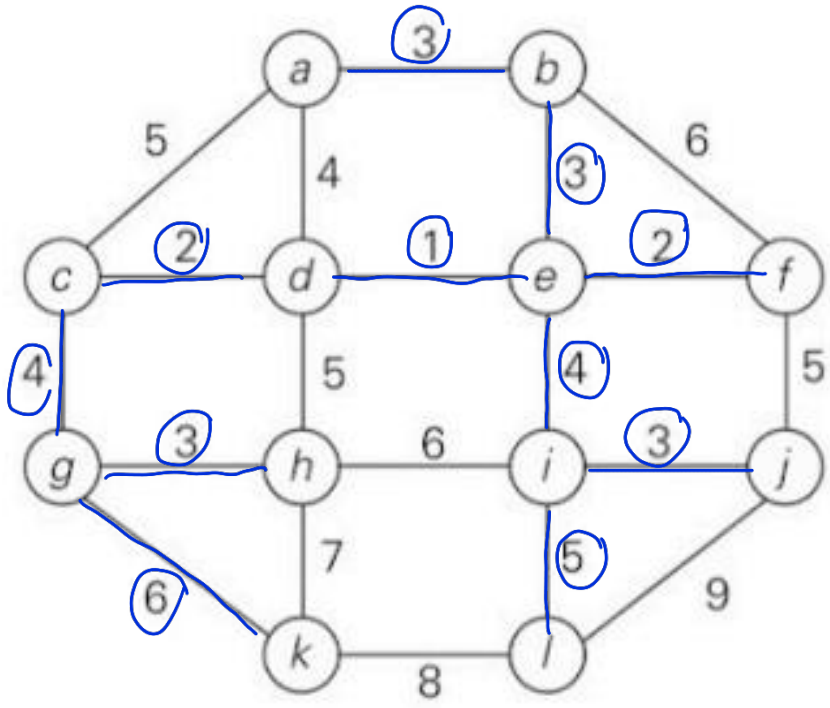
X = {s, 1 , 2} 12 + 14 = 26

X = {s, 1, 3} 13 + 9 + 20 = 45

X = {s, 1, 2, 4} 12 + 7 + 4 = 23

X = {s, 1, 3, 4} 20 + 4 = 24

What is the cost of minimum spanning tree of the graph below ? *



28

31

33

36

38

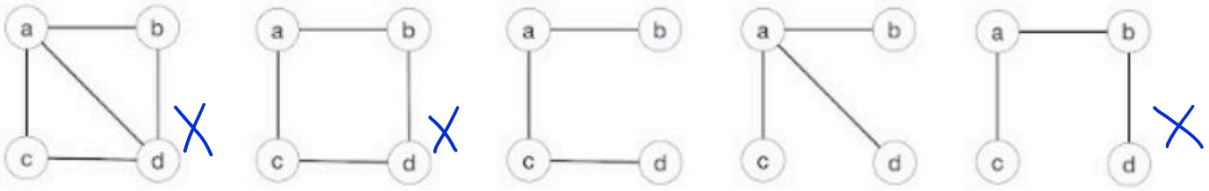
41

Given an adjacency matrix of graph G, which subgraphs are not spanning trees of G ? *

2 points

Adjacency matrix of G

	a	b	c	d
a	0	1	1	1
b	1	0	0	0 → 1 X
c	1	0	0	1
d	1	0	1	0



G_1 G_2 G_3 G_4 G_5

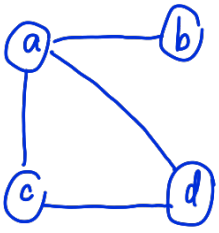
G1

G2

G3

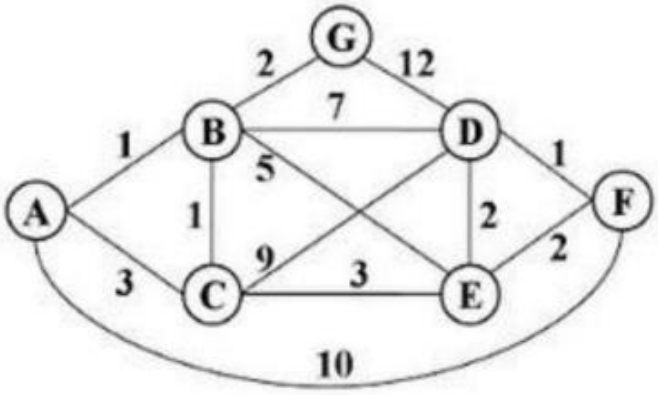
G4

G5



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Consider the following undirected, weighted graph:



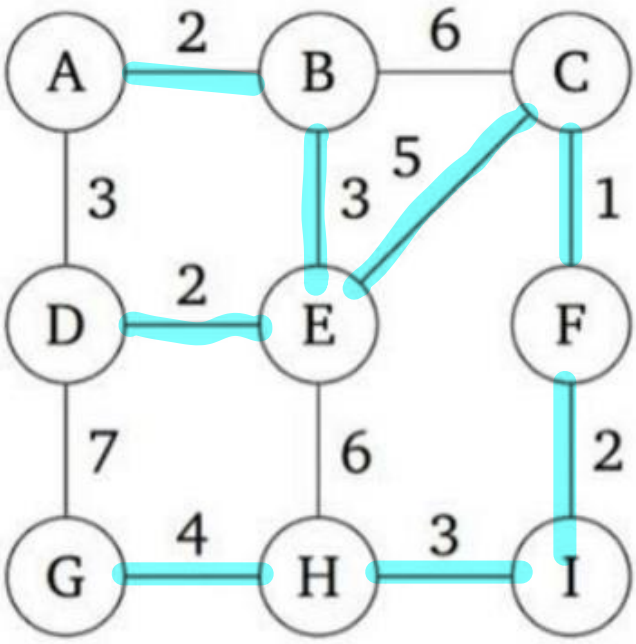
Given N_i $(N_{i-1}, cost)$ be the notation of vertices in the shortest path using Dijkstra's algorithm. If Dijkstra's algorithm is applied to calculate the single-shortest paths from A to every other vertex. Which vertex notation is incorrect?

- ○ G(B, 3)
- ○ D(E, 7)
- ◉ E(D, 5)
- ○ C(B, 2)

What problems are Floyd-Warshall's algorithm used to solve?

- ◉ All pair shortest path problems
- ○ Single Source shortest path problems
- ○ Network flow problems
- ○ Minimum spanning tree problems

Running Kruskal's algorithm on the graph below, which edge is not contained in the minimum spanning tree?



- BE
- HI
- AD
- CE

Which of the following greedy algorithms has an implementation that most resembles that of Dijkstra's algorithm?

- Huffman's algorithm
- Floyd-Warshall algorithm
- Prim's algorithm
- Kruskal's algorithm

What problems are Dijkstra's algorithm used to solve?

- (●) Single Source shortest path problems
- ( ) Minimum spanning tree problems
- ( ) Network flow problems
- ( ) All pair shortest path problems

What problems are Kruskal's algorithm used to solve?

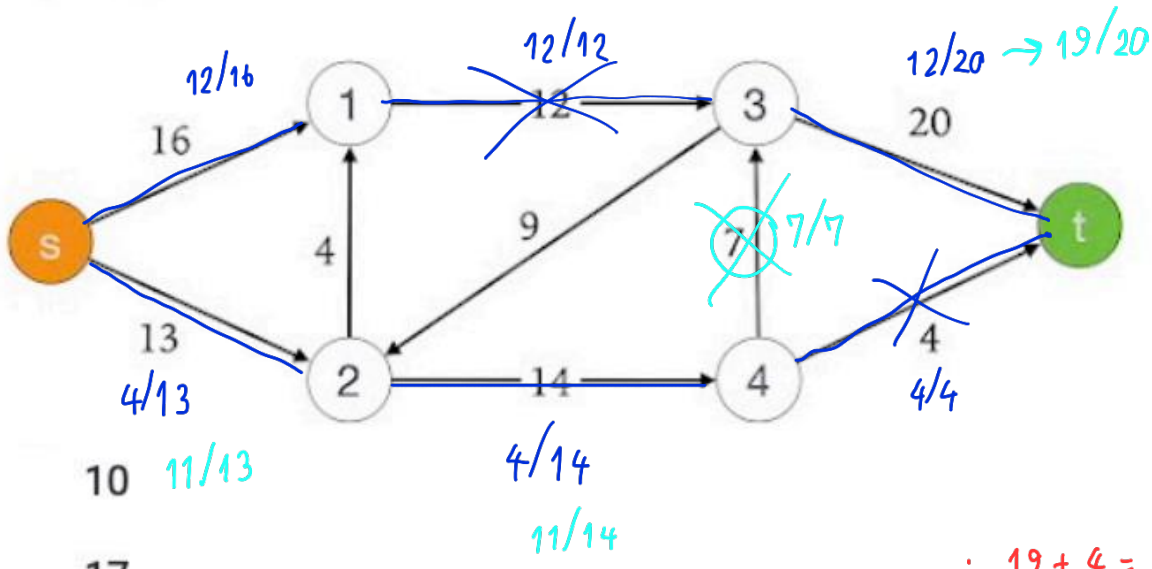
- ( ) Network flow problems
- ( ) Single Source shortest path problems
- ( ) All pair shortest path problems
- (●) Minimum spanning tree problems

When following Kruskal's algorithm, the greedy choice is to

- ( ) Add the edge of least cost to the forest so long as its addition does not create a cycle.
- ( ) Remove the vertex having greatest connection cost from the tree.
- ( ) Remove the edge of greatest cost from the graph so long as its removal does not disconnect the graph.
- (●) Add the vertex having least connection cost to the current tree.

มีโอกาสออก

For the following s-t flow network, what is the maximum s-t flow ? *



10

17

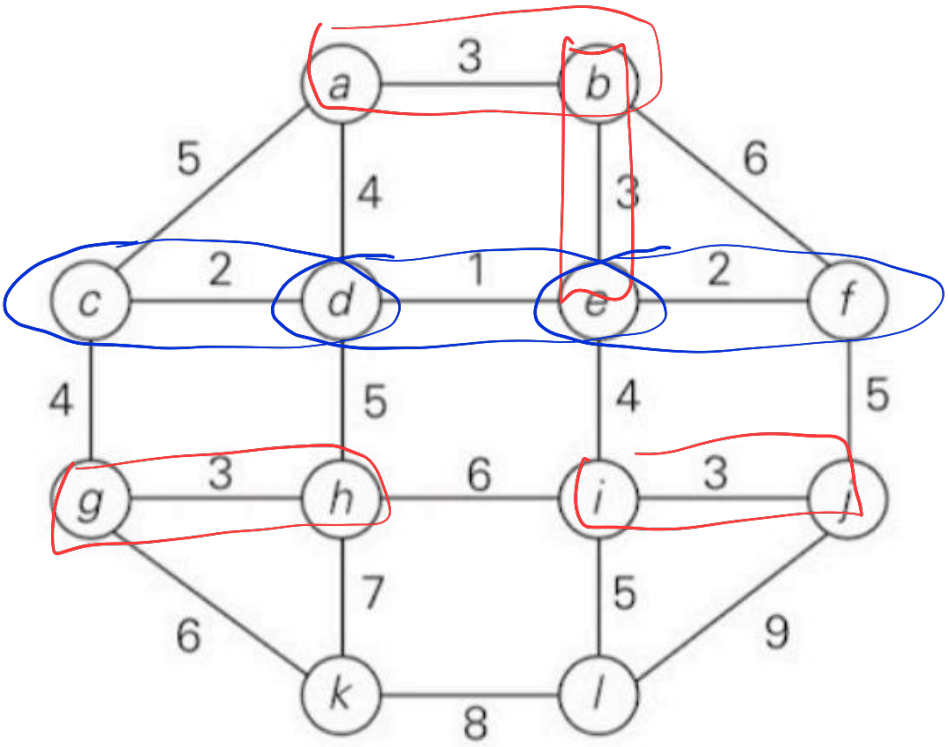
19

23

27

∴ 19 + 4 = 23

If Kruskral's algorithm is applied to the graph below, which edges could be added to the spanning tree in the fourth order? *



(a, b)

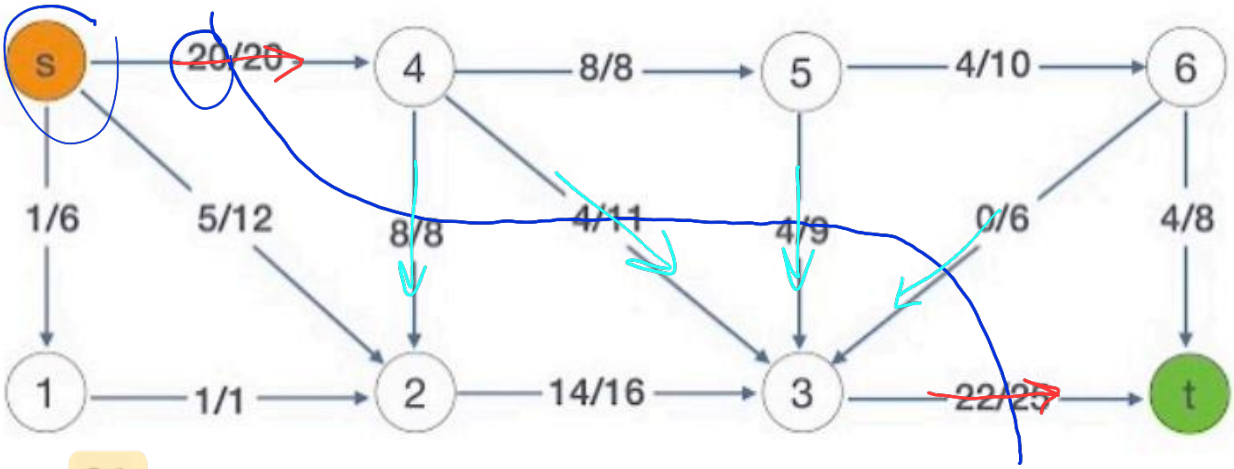
(c, d)

(e, f)

(d, a)

(h, g)

For the following s-t flow network, what are the cut capacity and the net flow across the cut with X = {s, 1, 2, 3} ? * 2 points



26

33

36

45

79

cut 20 + 22 - 8 - 4 - 4 - 0 = 26

flow 20 + 25 = 45